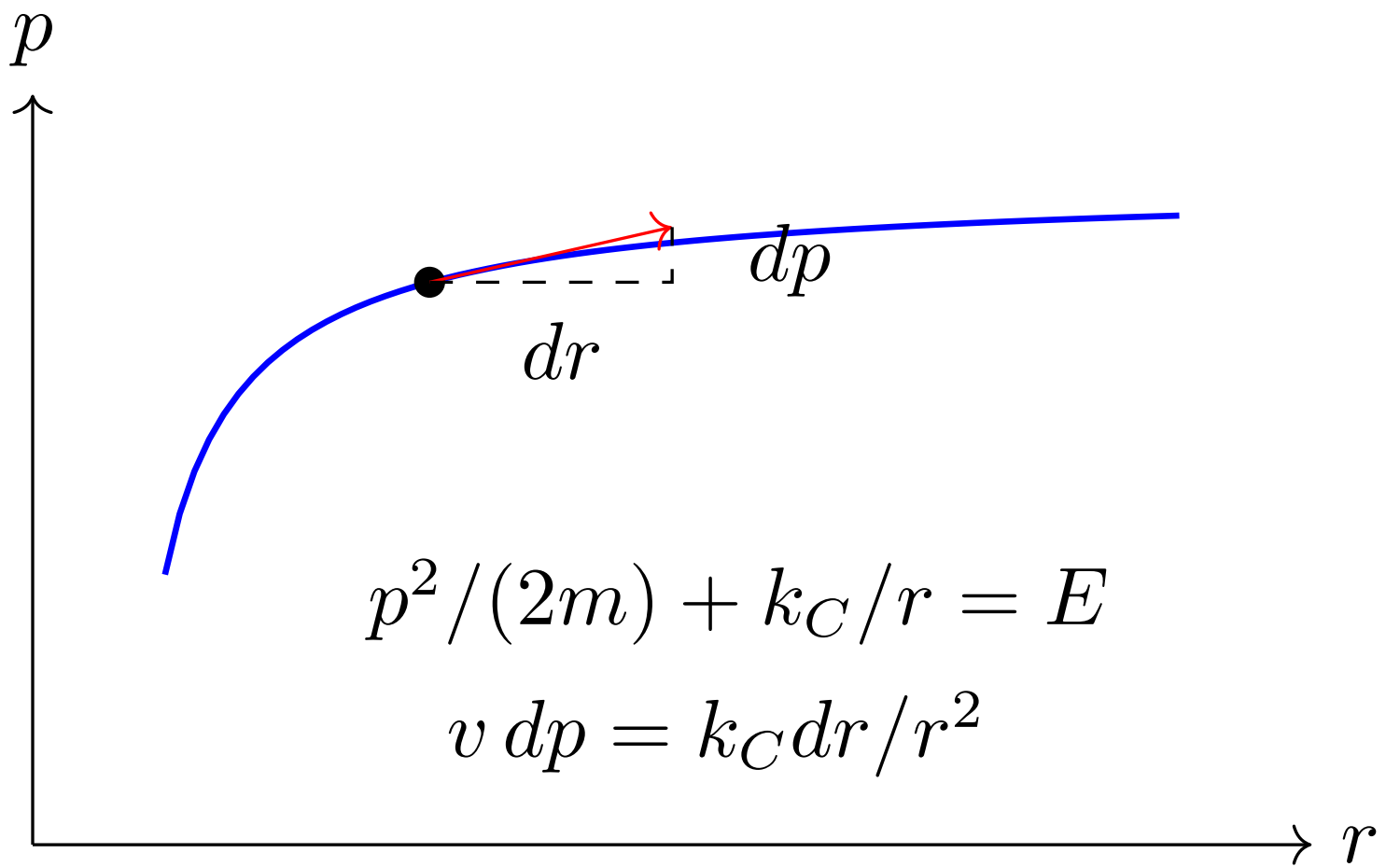

$$p^2/(2m) + k_C/r = E$$

$$v\,dp = k_C\,dr/r^2$$